

Indicator Removal

Sub-techniques (8)



Adversaries may selectively delete or modify artifacts generated to reduce indications of their presence and blend in with legitimate activity. Rather than broadly removing evidence, adversaries may target specific artifacts that appear anomalous or are likely to draw scrutiny, while leaving sufficient data intact to maintain the appearance of normal system behavior.

Artifacts such as command histories, log entries, or file metadata may be altered in ways that align with expected user or system activity. Location, format, and type of artifact (such as command or login history) are often platform-specific, allowing adversaries to tailor modifications that minimize suspicion.

These actions may not prevent detection entirely but can delay recognition of malicious activity or reduce the fidelity of alerts by making events appear benign or consistent with routine operations. Additionally, selectively removed or modified artifacts may still be recoverable through deeper forensic analysis, though their absence or alteration can complicate timeline reconstruction and attribution.

ID: T1070

Sub-techniques: [T1070.003](#), [T1070.004](#), [T1070.005](#), [T1070.006](#), [T1070.007](#), [T1070.008](#), [T1070.009](#), [T1070.010](#)

- ① **Tactic:** [Stealth](#)
- ① **Platforms:** Containers, ESXi, Linux, Network Devices, Office Suite, Windows, macOS

Contributors: Blake Strom, Microsoft 365 Defender; Brad Geesaman, @bradgeesaman; Ed Williams, Trustwave, SpiderLabs

Version: 3.0

Created: 31 May 2017

Last Modified: 12 May 2026

[Version Permalink](#)

Procedure Examples

ID	Name	Description
G1044	APT42	APT42 has cleared Chrome browser history. ^[1]
G1023	APT5	APT5 has used the THINBLOOD utility to clear SSL VPN log files located at <code>/home/runtime/logs</code> . ^{[2][3]}
S0239	Bankshot	Bankshot deletes all artifacts associated with the malware from the infected machine. ^[4]
S0089	BlackEnergy	BlackEnergy has removed the watermark associated with enabling the <code>TESTSIGNING</code> boot configuration option by removing the relevant strings in the <code>user32.dll.mui</code> of the system. ^[5]
S1161	BPFDoor	BPFDoor clears the file location <code>/proc/<PID>/environ</code> removing all environment variables for the process. ^[6]
S0527	CSPY Downloader	CSPY Downloader has the ability to remove values it writes to the Registry. ^[7]
C0029	Cutting Edge	During Cutting Edge , threat actors cleared logs to remove traces of their activity and restored compromised systems to a clean state to bypass manufacturer mitigations for CVE-2023-46805 and CVE-2024-21887. ^{[8][9]}
S0673	DarkWatchman	DarkWatchman can uninstall malicious components from the Registry, stop processes, and clear the browser history. ^[10]
S0695	Donut	Donut can erase file references to payloads in-memory after being reflectively loaded and executed. ^[11]
S1159	DUSTTRAP	DUSTTRAP restores the <code>.text</code> section of compromised DLLs after malicious code is loaded into memory and before the file is closed. ^[12]
S0568	EVILNUM	EVILNUM has a function called "DeleteLeftovers" to remove certain artifacts of the attack. ^[13]
S0696	Flagpro	Flagpro can close specific Windows Security and Internet Explorer dialog boxes to mask external connections. ^[14]
S1044	FunnyDream	FunnyDream has the ability to clean traces of malware deployment. ^[15]
S0697	HermeticWiper	HermeticWiper can disable pop-up information about folders and desktop items and delete Registry keys to hide malicious services. ^{[16][17]}
S1132	IPsec Helper	IPsec Helper can delete various registry keys related to its execution and use. ^[18]
S9029	IronWind	IronWind has used a .NET DLL named "exit-DN4-core.dll" to terminate malicious processes running on victim's systems. ^[19]
G0032	Lazarus Group	Lazarus Group has restored malicious KernelCallbackTable code to its original state after the process execution flow has been hijacked. ^[20]
S0449	Maze	Maze has used the "Wow64RevertWow64FsRedirection" function following attempts to delete the shadow volumes, in order to leave the system in the same state as it was prior to redirection. ^[21]
S0455	Metamorfo	Metamorfo has a command to delete a Registry key it uses, <code>\Software\Microsoft\Internet Explorer\notes</code> . ^[22]
S1135	MultiLayer Wiper	MultiLayer Wiper uses a batch script to clear file system cache memory via the <code>ProcessIdleTasks</code> export in <code>advapi32.dll</code> as an anti-analysis and anti-forensics technique. ^[23]
G0129	Mustang Panda	Mustang Panda has deleted registry keys that store data and maintained persistence. ^[24]

ID	Name	Description
S0448	Rising_Sun	Rising_Sun can clear a memory blog in the process by overwriting it with junk bytes. ^[28]
S1085	Sardonic	Sardonic has the ability to delete created WMI objects to evade detections. ^[29]
S0461	SDBbot	SDBbot has the ability to clean up and remove data structures from a compromised host. ^[30]
S0596	ShadowPad	ShadowPad has deleted arbitrary Registry values. ^[31]
S0589	Sibot	Sibot will delete an associated registry key if a certain server response is received. ^[32]
S0692	SILENTTRINITY	SILENTTRINITY can remove artifacts from the compromised host, including created Registry keys. ^[33]
C0024	SolarWinds Compromise	During the SolarWinds Compromise , APT29 temporarily replaced legitimate utilities with their own, executed their payload, and then restored the original file. ^[34]
S0603	Stuxnet	Stuxnet can delete OLE Automation and SQL stored procedures used to store malicious payloads. ^[35]
S0559	SUNBURST	SUNBURST removed HTTP proxy registry values to clean up traces of execution. ^[36]

Mitigations

ID	Mitigation	Description
M1041	Encrypt Sensitive Information	Obfuscate/encrypt event files locally and in transit to avoid giving feedback to an adversary.
M1029	Remote Data Storage	Automatically forward events to a log server or data repository to prevent conditions in which the adversary can locate and manipulate data on the local system. When possible, minimize time delay on event reporting to avoid prolonged storage on the local system.
M1022	Restrict File and Directory Permissions	Protect generated event files that are stored locally with proper permissions and authentication and limit opportunities for adversaries to increase privileges by preventing Privilege Escalation opportunities.

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0184	Behavioral Detection of Indicator Removal Across Platforms	AN0520	Monitors sequences involving deletion/modification of logs, registry keys, scheduled tasks, or prefetch files following suspicious process activity or elevated access escalation.
		AN0521	Detects deletion or overwriting of bash history, syslog, audit logs, and .ssh metadata following privilege elevation or suspicious process spawning.
		AN0522	Detects clearing of unified logs, deletion of plist files tied to persistence, and manipulation of Terminal history after initial execution.
		AN0523	Monitors tampering with audit logs, volumes, or mounted storage often used for side-channel logging (e.g., /var/log inside containers) post-compromise.
		AN0524	Tracks suspicious use of ESXi shell commands or PowerCLI to delete logs, rotate system files, or tamper with hostd/vpxa history.
		AN0525	Detects deletion or hiding of security-related mailboxes, audit mailboxes, and log data/log

References

1. [Rozmann, O., et al. \(2024, May 1\). Uncharmed: Untangling Iran's APT42 Operations. Retrieved October 9, 2024.](#)
2. [Perez, D. et al. \(2021, April 20\). Check Your Pulse: Suspected APT Actors Leverage Authentication Bypass Techniques and Pulse Secure Zero-Day. Retrieved February 5, 2024.](#)
3. [Perez, D. et al. \(2021, May 27\). Re-Checking Your Pulse: Updates on Chinese APT Actors Compromising Pulse Secure VPN Devices. Retrieved February 5, 2024.](#)
4. [US-CERT. \(2017, December 13\). Malware Analysis Report \(MAR\) - 10135536-B. Retrieved July 17, 2018.](#)
5. [F-Secure Labs. \(2014\). BlackEnergy & Quedagh: The convergence of crimeware and APT attacks. Retrieved March 24, 2016.](#)
6. [The Sandfly Security Team. \(2022, May 11\). BPFDoor - An Evasive Linux Backdoor Technical Analysis. Retrieved September 29, 2023.](#)
7. [Dahan, A. et al. \(2020, November 2\). Back to the Future: Inside the Kimsuky KGH Spyware Suite. Retrieved November 6, 2020.](#)
8. [Lin, M. et al. \(2024, January 31\). Cutting Edge, Part 2: Investigating Ivanti Connect Secure VPN Zero-Day Exploitation. Retrieved February 27, 2024.](#)
9. [Meltzer, M. et al. \(2024, January 10\). Active Exploitation of Two Zero-Day Vulnerabilities in Ivanti Connect Secure VPN. Retrieved February 27, 2024.](#)
10. [Smith, S., Stafford, M. \(2021, December 14\). DarkWatchman: A new evolution in fileless techniques. Retrieved January 10, 2022.](#)
11. [TheWover. \(2019, May 9\). donut. Retrieved March 25, 2022.](#)
12. [Mike Stokkel et al. \(2024, July 18\). APT41 Has Arisen From the DUST. Retrieved September 16, 2024.](#)
13. [Adamitis, D. \(2020, May 6\). Phantom in the Command Shell. Retrieved November 17, 2024.](#)
14. [Hada, H. \(2021, December 28\). Flagpro The new malware used by BlackTech. Retrieved March 25, 2022.](#)
15. [Vrabie, V. \(2020, November\). Dissecting a Chinese APT Targeting South Eastern Asian Government Institutions. Retrieved September 19, 2022.](#)
16. [Thomas, W. et al. \(2022, February 25\). CrowdStrike Falcon Protects from New Wiper Malware Used in Ukraine Cyberattacks. Retrieved March 25, 2022.](#)
17. [ESET. \(2022, March 1\). IsaacWiper and HermeticWizard: New wiper and worm targeting Ukraine. Retrieved April 10, 2022.](#)
18. [Amitai Ben & Shushan Ehrlich. \(2021, May\). From Wiper to Ransomware: The Evolution of Agrius. Retrieved May 21, 2024.](#)
19. [Check Point. \(2024, November 12\). Hamas-affiliated Threat Actor WIRTE Continues its Middle East Operations and Moves to Disruptive Activity. Retrieved April 20, 2026.](#)
20. [Saini, A. and Hossein, J. \(2022, January 27\). North Korea's Lazarus APT leverages Windows Update client, GitHub in latest campaign. Retrieved January 27, 2022.](#)
21. [Mundo, A. \(2020, March 26\). Ransomware Maze. Retrieved May 18, 2020.](#)
22. [Sierra, E., Iglesias, G.. \(2018, April 24\). Metamorfo Campaigns Targeting Brazilian Users. Retrieved July 30, 2020.](#)
23. [Or Chechik, Tom Fakterman, Daniel Frank & Assaf Dahan. \(2023, November 6\). Agonizing Serpens \(Aka Agrius\) Targeting the Israeli Higher Education and Tech Sectors. Retrieved May 22, 2024.](#)
24. [Alexandre Cote Cyr. \(2022, March 23\). Mustang Panda's Hodur: Old tricks, new Korplug variant. Retrieved September 9, 2025.](#)
25. [MSTIC. \(2021, December 6\). NICKEL targeting government organizations across Latin America and Europe. Retrieved March 18, 2022.](#)
26. [Axel F, Pierre T. \(2017, October 16\). Leviathan: Espionage actor spearphishes maritime and defense targets. Retrieved February 15, 2018.](#)
27. [Zhang, X. \(2024, November 8\). New Campaign Uses Remcos RAT to Exploit Victims. Retrieved April 16, 2026.](#)
28. [Sherstobitoff, R., Malhotra, A., et. al.. \(2018, December 18\). Operation Sharpshooter Campaign Targets Global Defense, Critical Infrastructure. Retrieved May 14, 2020.](#)
29. [Budaca, E., et al. \(2021, August 25\). FIN8 Threat Actor Goes Agile with New Sardonic Backdoor. Retrieved August 9, 2023.](#)
30. [Schwarz, D. et al. \(2019, October 16\). TA505 Distributes New SDBbot Remote Access Trojan with Get2 Downloader. Retrieved May 29, 2020.](#)
31. [Kaspersky Lab. \(2017, August\). ShadowPad: popular server management software hit in supply chain attack. Retrieved March 22, 2021.](#)
32. [Nafisi, R., Lelli, A. \(2021, March 4\). GoldMax, GoldFinder, and Sibot: Analyzing NOBELIUM's layered persistence. Retrieved March 8, 2021.](#)
33. [Salvati, M. \(2019, August 6\). SILENTTRINITY Modules. Retrieved March 24, 2022.](#)
34. [FireEye. \(2020, December 13\). Highly Evasive Attacker Leverages SolarWinds Supply Chain to Compromise Multiple Global Victims With SUNBURST Backdoor. Retrieved January 4, 2021.](#)
35. [Nicolas Falliere, Liam O Murchu, Eric Chien 2011, February W32.Stuxnet Dossier \(Version 1.4\) Retrieved November 17, 2024.](#)
36. [MSTIC, CDOC, 365 Defender Research Team. \(2021, January 20\). Deep dive into the Solorigate second-stage activation: From SUNBURST to TEARDROP and Raindrop. Retrieved January 22, 2021.](#)